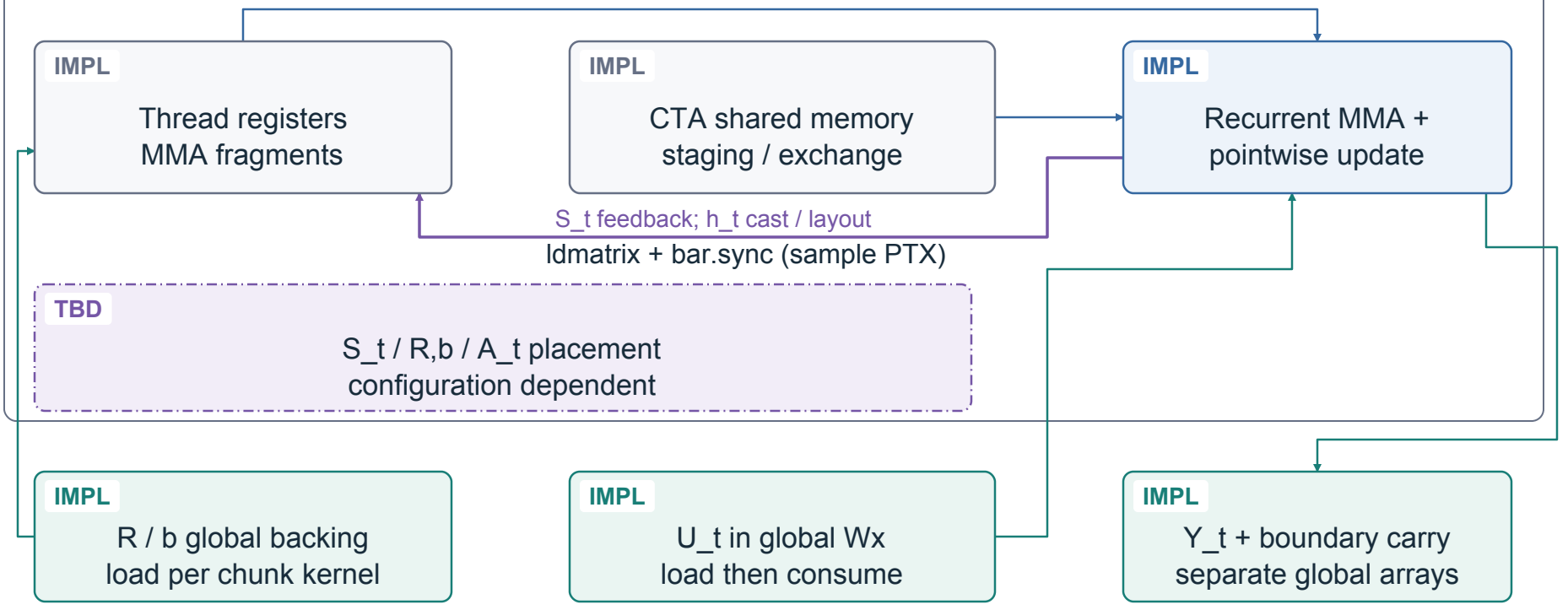
State residency: scope, storage and lifetime

Generic MMA path | one CTA / batch tile / head | exact object allocation: TBD

Execution scope: one CTA / batch tile / head

Storage spaces below are distinct from the execution scope.



Lifetime within one chunk (logical order; not to scale)

	init	t	t+1	finish
R / b	load once	reuse	reuse	release
S_t	initial S_0	update S_t	consume / update	FP32 carry
U_t	global Wx	ready -> use	new U_(t+1)	slots TBD
A_t	not live	dot -> update	new A_(t+1)	not live
Y_t	allocate output	full/hidden: Y_t	full/hidden: Y_(t+1)	last: final h
C_t	absent in this inference scope			

No CTA-to-SM ownership claim. Exact state layout / spills: TBD.
Borrowed prepared results must be consumed before the next run overwrites them.

State / feedback · Compute · Input / output · Sync / constraints · Storage / scope

IMPL: Implemented · **DOC: Documented** · **PROPOSED** · **TBD: Unknown**